

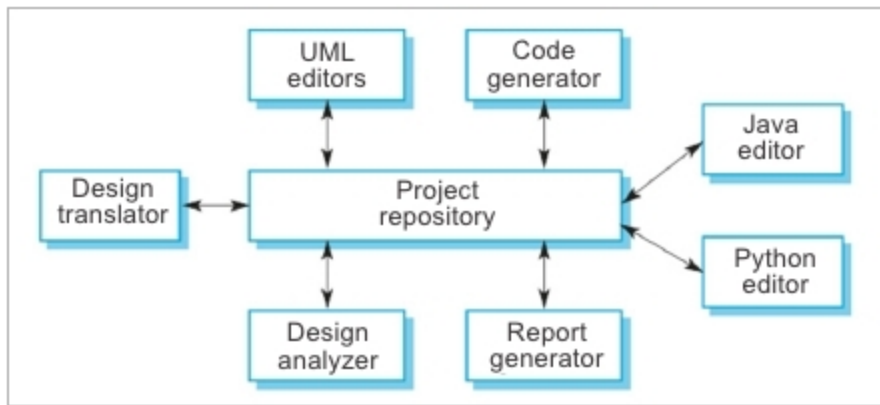


Figure 1: Repository architecture for an IDE (image source: *googleimages.com*)

building GUIs. It has also got a marketplace named the Eclipse Marketplace Client that gives its users access to a large treasure trove of information and plugins supplied by an expanding community of developers.

Languages supported: C++, C, Java, PHP, Perl, Python, Ruby and more.

Features:

1. Eclipse has got a plethora of package solutions that help it to support multiple programming languages.
2. It supports Java, as well as different IDE enhancements like hierarchical views of various nested projects with customisable perspectives.
3. It has got a task-focused interface that supports system-tray notifications as well.
4. Eclipse has an automated error reporting feature that automatically generates a descriptive error report whenever an error occurs.
5. There are multiple tooling options available in Eclipse for JEE projects.
6. Eclipse can be easily integrated with JUnit as well.

Drawbacks: All of us know that Eclipse is very versatile, but it has got multiple options that may be intimidating to newcomers. At times, it becomes challenging to add external plugins to Eclipse in order to utilise the features supported by that plugin.

NetBeans

This is another free and open source IDE that is ideal for editing any existing project or even starting a project from scratch. NetBeans provides a simple drag-and-drop interface that is equipped with many convenient project templates. It's primarily used to develop Java applications, but various packages or bundles that support other languages as well can be downloaded.

Languages supported: C, FORTRAN, C++, C++11, HTML 5, Java, PHP and more.

Features:

1. NetBeans provides an intuitive drag-and-drop interface that makes the software development process easier.

2. It has got both static and dynamic libraries that can be used for software development.
3. Multi-session GNU debuggers can be easily integrated with each other, providing code assistance.
4. NetBeans allows its users to develop software remotely.
5. It is compatible with Windows, Linux, OS X and Solaris.
6. It supports FORTRAN and Assembler files.
7. Supports Qt Toolkit.
8. It supports a number of compilers like Cygwin, Clang/LLVM, MinGW, GNU and Oracle Solaris Studio.

Drawbacks: Some NetBean's users have reported that it requires a lot of memory; so it is possible that it may perform sluggishly on machines without enough memory.

Code::Blocks

Code::Blocks is one of the popular open source IDE options available for developers working on Windows, Linux and Mac. It is a highly customisable IDE that performs consistently across all the platforms; so it is really great for developers who frequently switch their workspaces from one platform to another. The plug-in framework supported by Code::Blocks lets its users tweak it as per their needs and requirements.

Languages supported: FORTRAN, C and C++.

Features:

1. Code::Blocks is compatible with Windows, Mac and Linux.
2. It is developed using C++.
3. It supports an easy-to-navigate tabbed interface that includes a list of open files as well.
4. Code::Blocks requires no interpreted or proprietary languages.
5. It extends itself to support a wide range of custom-built and pre-built plug-ins.
6. Supports multiple compilers like MSVC++, GCC, Clang and more.
7. It provides users with a debugger that supports breakpoint.
8. Code::Blocks supports a text editor that provides syntax highlighting and code auto-completion.
9. Customisable with external tools.
10. It has got simple task management tools that are ideal for its new users.

Drawbacks: Although Code::Blocks comes with a list of features, it is a comparatively lightweight IDE. It is not suitable for larger projects. It is indeed a great tool for beginners, but advanced coders may find it a bit frustrating because of its limitations.

Aptana Studio3

Aptana Studio3 is possibly one of the most powerful open source IDEs available in the market. It is a massive